MIL-A-25546C(USAF)

18 June 1970
Superseding
MIL-A-25546B(USAF)
15 April 1960

MILITARY SPECIFICATION

AMPLIFIER, AUDIO FREQUENCY AM-944/AIC-13

1. SCOPE.

1.1 This specification covers one type of audio frequency amplifier designated Amplifier, Audio Frequency AM-944/AIC-13, which is one item of Public Address Set AN/AIC-13.

2. APPLICABLE DOCUMENTS.

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-E-5400	Electronic Equipment, Aircraft, General Specification For
MIL-C-8384	Connectors, Electrical, (Molded Body) and Accessories
MIL-S-19500	Semiconductor Devices, General Specification For

STANDARDS

Military

MIL-STD-461	Electromagnetic Interference Characteristics Requirement for Equipment
MIL-STD-704	Electronic Power Aircraft, Characteristics and Utilization
MIL-STD-781	Reliability Tests Exponential Distribution

FSC 5831

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:40Z
Check the source to verify that this is the current version before use.

DRAWINGS

Air Force

60D12544	Wiring Diagram AM-944/AIC-13 Audio Frequency Amplifier, Schematic
54D12570	Mounting MT-1424/AIC-10, Assembly Of
56J12718	Installation Information, Wiring Diagram, Public Address Set AN/AIC-13
56D12719	Amplifier, Audio Frequency AM-944/AIC-13 Outline Dimensions Of
56D12721	Wiring Diagram, AM-944/AIC-13 Audio Frequency Amplifier, Test Circuit, Schematic

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS.

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 General Specification. The requirement of MIL-E-5400 for Class 2 equipment apply as requirements of this specification with the exceptions and additions called out herein. When the two conflict, this specification shall govern.

3.2.1 Substitution of Material. If a manufacturer, either prior to or during production, desires to substitute another material or process for one previously approved or specified, he shall submit to the procuring activity a statement to that effect. The statement shall describe the proposed substitution and the reason therefor, together with evidence to show that the substitution is satisfactory. At the discretion of the procuring activity, test samples or additional data may be required to prove the adequacy of the proposed substitution.

3.2.2 Connector. The external electrical connector shall be compatible with the receptacle on MT-1424/AIC-10 and meet the requirements of AF Drawing 56D12719 and MIL-C-8384.

3.2.3 Transistors. The transistors shall meet the requirements of MIL-S-19500.

3.2.4 Electromagnetic Interference. Electromagnetic interference control testing shall be in accordance with applicable requirements of MIL-STD-461, Table I, for Class IC equipment.

3.2.5 Electrical Circuit. The functional circuit schematic of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:40Z
Check the source to verify that this is the current version before use.

AM-944/AIC-13 amplifier as shown in Drawing 60D12544 shall be used as a reference to determine the general circuitry and components of the amplifier. The contractor has the option to deviate from the circuit given, provided the functional characteristics of this specification are met.

3.2.6 Circuit Label Schematic. The actual circuits employed in the design of the AM-944/AIC-13 amplifier shall be shown by a schematic drawing similar to Drawing 60D12544. The circuit schematic label shall be firmly secured to the inside surface of the cover. Word identification of the terminals shall be included. A sketch of the proposed label shall be submitted to the procuring activity prior to the fabrication of the proposed circuit label.

3.3 Design and Construction.

3.3.1 Reliability. The reliability requirements of MIL-STD-781 shall be met as specified herein. The specified contract mean-time-between-failure (MTBF) shall not be less than 6000 hours.

3.3.2 Service Conditions (Electrical). The equipment shall be designed to operate from a power source with electrical characteristics conforming to the d-c power requirements of MIL-STD-704 and Category B, Table II steady state limits.

3.3.3 Mounting Provisions. The amplifier shall be designed to mount properly and securely with Mounting MT-1424/AIC-10. This mounting is detailed on Drawing 54D12570. Mounting MT-1424/AIC-10 is not a part of Amplifier, Audio Frequency AM-944/AIC-13.

3.4 Dimensions. The dimensions of the amplifier shall be in accordance with Drawing 56D12719.

3.4.1 Weight. The weight of the amplifier shall not exceed 5 pounds.

3.5 Performance.

3.5.1 Input Impedance. Two input impedances shall be provided. The low impedance shall be 150 ohms and the bridging impedance shall be 2000 ohms. The required impedance shall be within plus or minus 15 percent over the 300 to 5000 Hz frequency range.

3.5.2 Output Impedance. The output transformer shall provide output impedances of 4, 8, and 16 ohms.

3.5.3 Input-Output Characteristic.

3.5.3.1 Linearity. The input-output characteristic at 1000 Hz shall be in

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:40Z
Check the source to verify that this is the current version before use.

accordance with Table I.

TABLE I

Input Volts to Test Set		16-ohm output	8-ohm output	4-ohm output
150-ohm input	Bridging input	volts ± 2 db	volts ± 2 db	volts ± 2 db
0.056	0.40	1.8	1.27	0.9
0.18	1.25	5.8	4.0	2.9
0.56	4.00	18.0	12.7	9.0

3.5.3.2 Clipping. The output of the amplifier shall be symmetrically clipped for input signals greater than those required to produce an unclipped power output of 20 watts. This requirement shall apply for all signal inputs up to those required to produce 12 db of peak clipping.

3.5.4 Frequency Response Characteristic. The frequency response characteristic shall be within the limits of Table II.

TABLE II

Frequency (Hz)	Output (db re 1000 Hz output)	
	Min	Max
300	-2.0	+1
1000	0	0
3000	-2.0	+2.0
5000	-4.0	+2.0

3.5.5 Distortion. The total harmonic distortion present in the output of the amplifier at any frequency in the 300 to 5000 Hz band inclusive shall not exceed the values specified in Table III.

TABLE III

Power Output	Percent Distortion		
(watts)	16-ohms output	8-ohms output	4-ohms output
¼	5	5	5
½	5	5	5
1	5	5	5
5	5	5	5
10	6	6	7
15	8	9	9
20	13	14	15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:40Z
Check the source to verify that this is the current version before use.

3.5.6 Power Drain. The power drain of the amplifier shall not exceed 2.5 amperes at 27.5 volts d-c when the amplifier is delivering 20 watts to its load with both relays energized.

3.5.7 Phasing. The phase relationship of the input-to-output shall be such that when an instantaneous voltage at the "output high" terminal is positive with respect to ground, the instantaneous voltage at the "input high" is 180 degrees ± 15 degrees with respect to ground.

3.5.8 Operation. The amplifier shall provide satisfactory operation and shall not adversely affect the performance of any other component of Public Address Set AN/AIC-13 when interconnected with, and operated as a part of, that equipment. The amplifier shall be electrically and physically compatible with Control C-1614/AIC-13 and Loudspeaker LS-211/AIC-13. A typical hookup is shown on Drawing 56J12718.

3.6 Output Variation. The output of the amplifier shall not vary more than plus or minus 4.0 decibels from the output at room temperature when the amplifier is subjected to any of the service conditions (operating) described herein.

3.7 Government-Loaned Property. When specified in the contract or purchase order, the following items will be furnished by the Government on loan to the contractor upon his request:

Item Number	Item Description	Quantity
1	Amplifier, Audio Frequency AM-944/AIC-13	1
2	Control, Public Address Set C-1614/AIC-13	1
3	Loudspeaker, Magnetic LS-211/AIC-13	1

4. QUALITY ASSURANCE PROVISIONS.

4.1 Classification of Tests. The inspection and testing of AM Amplifier AM-944/AIC-13 shall be classified as follows:

a. Preproduction tests . . . . . See 4.3

b. Acceptance tests . . . . . See 4.4

4.2 Test Conditions. Unless otherwise specified, the amplifiers shall be under the following conditions:

a. Temperature: Room ambient plus 15°C to plus 35°C

b. Pressure: Normal atmospheric

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:40Z
Check the source to verify that this is the current version before use.

c. Humidity: Room ambient, up to 90 percent relative humidity.

d. Primary voltage supply: 27.5 volts d-c ± 1.0 volt.

4.3 Preproduction Tests. Preproduction tests shall be performed by the contractor in accordance with the following:

4.3.1 Preproduction Test Samples.

4.3.1.1 Contractor Test Samples. When specified in the contract a minimum of 10 each AF Amplifier AM-944/AIC-13 shall be subjected by the contractor to the preproduction tests specified herein. Seven (7) of these samples shall be used for reliability testing only.

4.3.1.2 Government Test Samples. When specified in the contract, the 3 each amplifiers not used above in reliability testing shall be submitted to the procuring activity. The sample amplifiers shall be accompanied by adequate identifying information including the Government order number involved ("File Number" in contract), if any, name and address of prime contractor, and name and address of manufacturer. The samples and data will be used by the procuring activity for:

a. Review of the mechanical design.

b. Performance of any tests, specified herein, which may be deemed necessary after a review of the contractor's report.

c. Government reference purposes.

4.3.2 Test Routine. The equipment shall be subjected to the tests listed in Table IV in the quantities shown. The order of conducting these tests need not be in the order listed.

TABLE IV

Inspection or Test	Paragraph or Requirement	Pre-Production	Individual	Sampling
Examination of Product	4.5.1	3	All	All
Electrical Tests	4.5.2			
Input-output Characteristic test	4.5.3		All	
Linearity	3.5.3.1	3	All	
Clipping	3.5.3.2	3		All
Frequency response characteristic	3.5.4	3		All
Frequency response test	4.5.4		All	

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:40Z
Check the source to verify that this is the current version before use.

Distortion	3.5.5	3		All
Distortion test	4.5.5		All	
Power drain	3.5.6	3		All
Phasing	3.5.7	3		
Talk-listen test	4.5.6	3	All	All
Operation	3.5.8	3		
Electromagnetic				
Interference Test	3.2.4	3		
Output Variation	3.6	3		
Reliability	3.3.1			
Reliability Test	4.5.8	7		
Environmental Tests	4.5.7			
Temperature Altitude		2		All
Vibration		2		
Shock		2		
Humidity		2		All
Salt Spray		2		All
Temperature Shock		2		
Sand and Dust		2		
Fungus		2		All

Note: Where a paragraph in Section 3 is referenced above, the test procedure shall consist of determining complete compliance with the requirements stated in that paragraph.

4.4 Acceptance Tests. Acceptance tests shall consist of:

a. Individual tests

b. Sampling Tests

4.4.1 Individual Tests. Each amplifier shall be subjected to the individual tests listed in Table IV.

4.4.2 Sampling Tests. Unless otherwise specified, sampling tests shall be conducted on at least one out of every 100 amplifiers on order. The samples shall be selected at random from quantities of amplifiers which have passed the individual tests. The samples shall be subjected to the sampling tests listed in Table IV in the quantities shown.

4.4.2.1 Acceptance Criteria. When one or more of the selected amplifiers fails to meet the specified requirements, acceptance of all items shall be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated. A failure analysis will be forwarded to the Procuring Activity.

4.5 Test Procedures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:40Z
Check the source to verify that this is the current version before use.

4.5.1 Examination of Product. The amplifier shall be examined to determine compliance with the requirements of this specification for materials, workmanship, dimensions, markings, weight, and finish. Each amplifier shall be installed into a Mounting MT-1424/AIC-10 to determine proper mating of connectors, guide pins, and retaining screws. The mounting shall have been checked to determine compliance with all dimensions on Drawing 54D12570.

4.5.2 Electrical Tests. The amplifier shall be electrically tested to determine compliance with this specification using a typical test circuit as shown on Drawing 56D12721.

4.5.3 Input-Output Characteristic Test. The amplifier output shall be measured at 1000 Hz with an input of 0.18 volt to the test circuit for the 150-Ohm input tap and 1.25 volts to the test circuit for the bridging input tap. This test shall be performed using the 16-ohm output impedance tap.

4.5.4 Frequency Response Test. The frequency response shall be measured using the 4-ohm output tap at 300 and 5000 Hz with an input of 0.18 volt to the test circuit of the 150-ohm input circuit.

4.5.5 Distortion Test. The distortion at 300 Hz shall be measured when the amplifier is developing 20 watts across load resistors of 16, 8, and 4 ohms respectively.

4.5.6 Talk-Listen Test. The amplifier shall be given a "talk-listen" test by connecting a speaker to one of the output taps on the amplifier and applying enough voice signal to the amplifier input to give a comfortable output from the speaker.

4.5.7 Environmental Tests. The amplifier shall be tested for conformance with the following tests of military specification MIL-E-5272:

Temperature-Altitude	Procedure I
Vibration	Procedure XII
Shock	Procedure V
Humidity	Procedure I
Salt Spray	Procedure I
Temperature Shock	Procedure I
Sand and Dust	Procedure I
Fungus	Procedure I

4.5.7.1 Criteria for Failure. The inability of an amplifier immediately after each environmental test to meet, within 15 percent, the requirements of the individual tests of Table IV shall constitute a failure.

4.5.8 Reliability Preproduction Test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:40Z
Check the source to verify that this is the current version before use.

4.5.8.1 Reliability Test. Reliability test as required by Paragraph 4.3.2 shall be conducted on Test Level F from MIL-STD-781.

4.5.8.2 Temperature Cycle. Temperature cycling shall be Standard Method of Temperature Cycling as specified in MIL-STD-781. The amplifier shall be maintained at the upper test level temperature for two (2) hours.

4.5.8.3 Specific Duty Cycle. Specific duty cycle shall be in accordance with MIL-STD-781.

4.5.8.4 Voltage Cycling. Voltage cycling shall be in accordance with MIL-STD-781.

4.5.8.5 Equipment Operating Parameters. The operating parameters shall be as specified in the individual tests of Table IV. Failure to meet the minimum requirements specified shall constitute a failure. Measurement of operating parameters shall be in accordance with MIL-STD-781.

4.5.8.6 Test Plan. Test Plan V is to be performed in accordance with MIL-STD-781 Table 2 with θ_0 = 6000 and θ_1 = 2000.

4.5.8.7 Test Option. The contractor has the option of running the reliability tests simultaneously with the other required preproduction tests with the stipulation that if any change is found needed by the other tests the reliability test must be restarted with zero time. If a failure occurs it will be analyzed, repaired and replaced on test.

5. PREPARATION FOR DELIVERY.

5.1 General. Preparation for delivery shall be in accordance with provisions specified in contract.

5.2 Shipment Marking. Shipment marking nomenclature shall be Amplifier, Audio Frequency AM-944/AIC-13.

6. NOTES.

6.1 Intended Use. Amplifier, Audio Frequency AM-944/AIC-13 is a power amplifier used to provide communication between stations within and the ground crew outside the aircraft during maintenance and loading. It is also used to brief and entertain passengers on long flights.

6.2 Ordering Data. Invitation for bids, contract, and purchase order will specify that 10 each AF Amplifier AM-944/AIC-13 shall be required as preproduction tests to determine compliance with the requirements of this document and that 3 amplifiers AM-944/AIC-13 not used in reliability testing shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:40Z
Check the source to verify that this is the current version before use.

be submitted to the procuring activity. The invitation for bids and the contract shall specify the point of inspection for these tests.

6.3 Changes From Previous Issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extentiveness of the changes.

Custodian:

Air Force - 84

Preparing Activity:

Air Force - 84

Project Number:

5831-F010

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:40Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE. AF-WP-O-MAY 64 505

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:40Z
Check the source to verify that this is the current version before use.